POSTGRESQL EXTENSIONS AS IMAGE VOLUMES

Decoupling extensions from the operand image with CloudNativePG

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EDB POSTGRES AI

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SPEAKER

ABOUT ME

GABRIELE ROSSINI

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APPRENTICE SPEAKER

I build Postgres observability for Kubernetes and traditional infrastructure.

Ten years from full-stack to backend and telemetry.

I lead teams and mentor engineers.

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ABOUT EDB

EDB BUILDS POSTGRES AND GAVE CLOUDNATIVEPG TO THE CNCF

WHY EDB

Largest commercial Postgres contributor

Ranked first in the Postgres Vitality Index, which EDB publishes.

Writes the code it supports

Break/fix patches for the database, under 24×7 contractual SLAs.

Enterprise security

Transparent Data Encryption, not available in community Postgres.

Certified on Red Hat OpenShift

EDB Postgres for Kubernetes is a certified OpenShift operator.

FROM EDB TO THE CNCF

2019 Cloud-native Postgres initiative starts at 2ndQuadrant.

2020 EDB acquires 2ndQuadrant and keeps investing.

2022 Open sourced as CloudNativePG under Apache 2.0.

2025 CNCF Sandbox. EDB transfers the copyright to the CNCF.

2026 CNCF Incubating since 2 November.

Sources: Postgres Vitality Index and EDB's CloudNativePG history (enterprisedb.com); CNCF project page (cncf.io/projects/cloudnativepg).

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CLOUDNATIVEPG IS A PRODUCTION-GRADE POSTGRES OPERATOR

CloudNativePG

CNCF Incubating project

Apache 2.0, open governance, 9,500+ GitHub stars.

The whole lifecycle in one Cluster resource

High availability, backups, point-in-time recovery, rolling updates.

High availability through the Kubernetes API

No Patroni or other external failover manager.

Public production adopters

IBM, Google Cloud, Microsoft Azure, GEICO, Tesla, Ericsson.

Sources: ADOPTERS.md and GitHub cloudnative-pg/cloudnative-pg (stars as of 6 Nov 2026); cncf.io/projects/cloudnativepg.

PROJECT SITE

cloudnative-pg.io

Docs, releases and community

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EVERY EXTENSION CHANGE REBUILDS THE POSTGRES IMAGE

TODAY

01 **Image size** The operand image carries PostgreSQL and every extension binary.

02 **Rebuild cadence** A new base image or extension version means a rebuild, a rescan and a rollout.

03 **Supply chain** You build and scan an image that upstream does not publish.

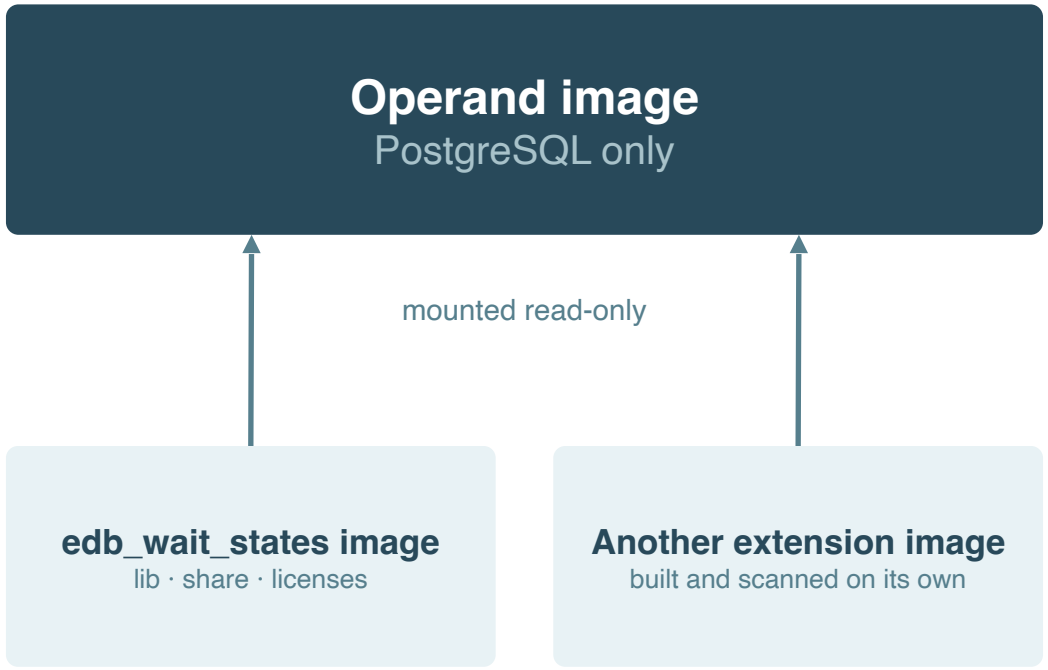
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EXTENSIONS MOVE OUT OF THE OPERAND IMAGE

TODAY

Custom operand image
PostgreSQL plus the extension binaries, built together

WITH IMAGE VOLUMES



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POSTGRESQL 18 FINDS EXTENSIONS OUTSIDE ITS OWN DIRECTORY

SHARED LIBRARIES

dynamic_library_path

Existing setting

Where the server looks for the shared libraries an extension loads.

CONTROL AND SQL FILES

extension_control_path

New in PostgreSQL 18

Where the server looks for an extension's control and SQL files, outside its compiled-in directory.

```
extension_control_path = '$system:/extensions/edb-wait-states/share'
dynamic_library_path   = '$libdir:/extensions/edb-wait-states/lib'
```

Values as set by CloudNativePG in this demo. $system and $libdir keep the built-in locations.

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KUBERNETES CAN MOUNT AN OCI IMAGE AS A READ-ONLY VOLUME

WHAT

An OCI image mounted read-only into a pod, as a volume.

WHEN

On by default from Kubernetes 1.35. On 1.33 and 1.34, enable the ImageVolume feature gate.

NEEDS

A runtime that supports it: containerd 2.1 or later, or CRI-O 1.31 or later.

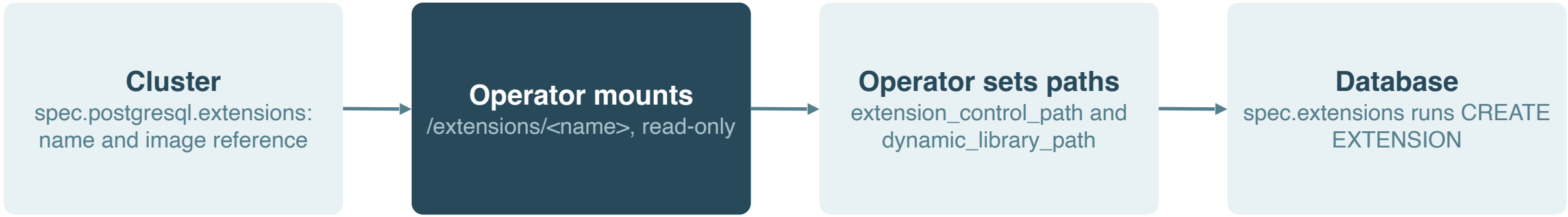
WHERE WE RAN IT

kind kindest/node v1.35.8, containerd 2.3.4

OpenShift 4.22.2, Kubernetes 1.35.5, CRI-O 1.35.4

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CLOUDNATIVEPG WIRES THE MOUNT INTO THE SETTINGS

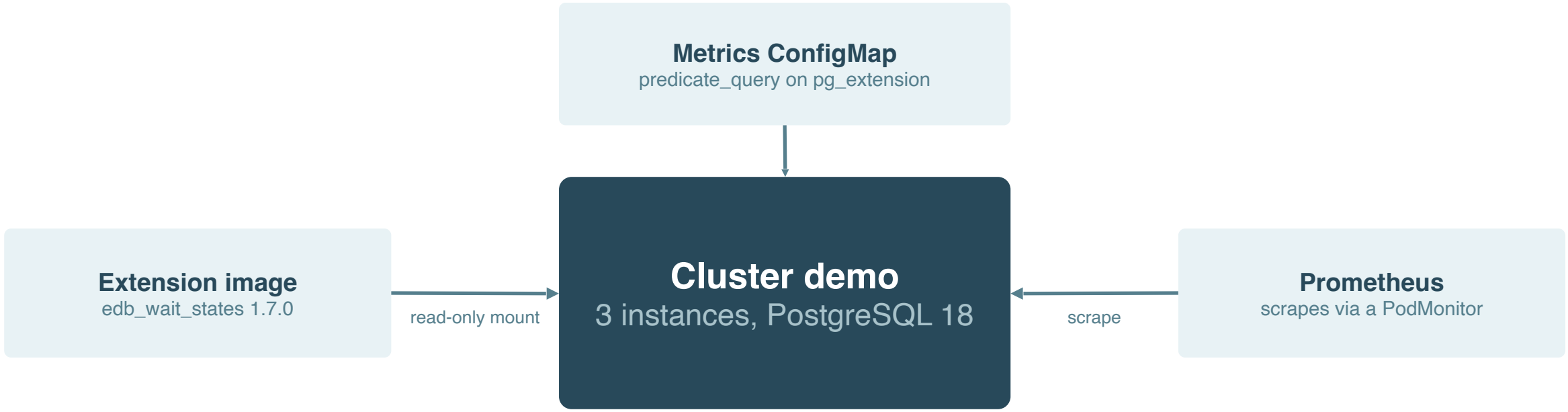


Extension management lives in the Cluster and Database resources, so it goes through the same review and GitOps flow as the rest of the configuration.

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THE METRIC APPEARS ONLY WHEN THE EXTENSION EXISTS



The same manifests run on kind and on OpenShift 4.22. Only the extension image reference differs.

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THE CHANGE IS ONE DIFF

```
spec:
  postgresql:
    shared_preload_libraries:
      - edb_wait_states
    extensions:
      - name: edb-wait-states
        image:
          reference: <extension image>
          pullPolicy: IfNotPresent
```

01 **Preload the library**

shared_preload_libraries

02 **Mount the image**

extensions: name and image reference

03 **Create the extension**

A Database resource, declaratively

IMAGE BUILD The extension image is built separately from the PostgreSQL image, from the EDB repository (EDB subscription token).

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LIVE DEMO

Baseline, diff, rolling update, wiring, Database, metric

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WRAP-UP

TAKEAWAYS

01

A minimal PostgreSQL image

Extensions ship as their own small images, built and scanned separately.

02

Two upstream features

PostgreSQL 18 and Kubernetes ImageVolume provide the mechanism. CloudNativePG wires them together.

03

Part of the Cluster

Adding or removing an extension is a change to the Cluster and Database resources.

THE POINT Adding an extension is a YAML diff, not a rebuild of the PostgreSQL image.

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BEFORE YOU ADOPT IT

LIMITS

01 PostgreSQL 18 or later

extension_control_path does not exist before 18.

02 Kubernetes ImageVolume

On by default from 1.35, a feature gate on 1.33 and 1.34, and the runtime must support it.

03 Matching images

The extension image must match the cluster's PostgreSQL major version, distribution and CPU architecture.

04 Restarts

A preloaded extension restarts every instance. By default the primary restarts in place.

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THANK YOU

Open source (MIT). edb_wait_states needs an EDB subscription token.

DEMO REPOSITORY

github.com/gabrosys/cnpg-imagevolume-extensions

LINKEDIN

linkedin.com/in/gabriele-rossini

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BACKUP

THE OPERATOR MOUNTED THE IMAGE AND SET TWO PATHS

```
/extensions/edb-wait-states
  lib/  licenses/  share/

extension_control_path
  $system:/extensions/edb-wait-states/share
dynamic_library_path
  $libdir:/extensions/edb-wait-states/lib

shared_preload_libraries
  edb_wait_states
```

58 s

Rolling update with 3 instances. The primary restarted in place.

88,728 B

Extension image layers, compressed, arm64.

Rolling update: measured 2026-11-01 on OpenShift 4.22.2 with CloudNativePG 1.30.1. Image size: 2026-11-04, local build. Other tools report other sizes.

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BACKUP

WHAT ELSE THE SAME MECHANISM COVERS

CATALOGS

Image catalogs

Extension images can be referenced from image catalogs, available since CloudNativePG 1.29.

COMMUNITY IMAGES

postgres-extensions-containers

A community project that ships pgvector, PostGIS, pgaudit and more as separate images.

UPGRADES

The version field

The version field in the Database resource drives the extension upgrade path.

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